

GEORGIA CAMPBELL (SHE/HER)

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EDUCATION

Mechanical and Biomedical Engineering

Year 4

McMaster University

September 2022 – April 2028

Hamilton, ON

RESEARCH EXPERIENCE

Computational Biomechanics Research Project

McMaster University

January 2026 – Present

Hamilton, ON

- Designed and conducted an ethics-approved computational biomechanics study using smartphone-based, markerless motion capture (OpenCap) to quantify 3D lower-extremity kinematics during an inside-of-the-foot soccer kick.
- Performed multi-view camera calibration, synchronized data collection, and OpenSim-based inverse kinematics to extract hip, knee, ankle, and pelvis kinematics, with emphasis on dominant versus non-dominant limb asymmetries.
- Interpreted kinematic outputs within known validation error bounds, discussing injury relevance, methodological limitations, and the feasibility of accessible field-based biomechanical analysis.

McMaster Injury Biomechanics Laboratory Undergraduate Research Student

McMaster University

September 2025 – Present

Hamilton, ON

- Designed and prototyped a TPU–silicone composite soft-tissue surrogate for anthropomorphic test devices (ATDs), including computer-aided design (CAD) development of interlocking geometries and casting molds.
- Evaluated candidate materials and planned ASTM-based compression and impact testing to validate biofidelity and durability.
- Conducted two 10-minute presentations for Mechanical Engineering faculty representatives and documented findings in a formalized report format.

Biochemistry and Biomedical Science (BBS) Summer Scholar

McMaster University

May 2023 – August 2023

Hamilton, ON

- Conducted molecular cloning and protein purification of MamJ, critical for magnetotactic bacteria and applied gel electrophoresis, polymerase chain reactions (PCR), and Bradford Assay techniques for validation.
- Created transformed plasmid vectors using Gibson Assembly and performed bacterial transformations.
- Delivered a professional 10-minute presentation of research findings to faculty and peers.

WORK EXPERIENCE

Instructional Teaching Assistant

McMaster University

September 2023 – Present

Hamilton, ON

- Led weekly tutorials and discussions to reinforce engineering design principles, foster collaboration, and improve problem-solving skills.
- Evaluated student projects through design reviews and grading, delivering constructive feedback to strengthen technical execution and design rigour.

Lab Teaching Assistant

McMaster University

September 2023 – Present

Hamilton, ON

- Mentored and supported 30+ undergraduate students in developing Wetlab techniques, communicating complex technical and biological concepts related to the clinical and socioeconomic impacts of diseases.
- Oversaw and instructed fundamental cell culturing, molecular cloning, and biological imaging techniques.

Program Development Intern

Hotchkiss Brain Institute

May 2025 – August 2025

Calgary, AB

- Developed and managed essential resource systems by implementing a new office inventory platform and compiling 3 comprehensive data banks supporting over 200 neuroscience researchers.
- Supported program logistics and event coordination for 2+ major institutional and international events, including HBI's Annual Research Day (300+ attendees) and the G7 Brain Economy Summit.

Biomedical Systems Co-op

Victoria Hand Project

May 2024 – August 2024

Victoria, BC

- Conducted safety tests and technical reports on new device components to verify design integrity.
- Modified 5+ upper-limb prosthetic device components, using SolidWorks and Fusion 360 to reduce assembly time and improve product functionality.
- Partnered with engineers, clinicians, and suppliers to resolve manufacturing challenges, manage material orders/vendor quotes, and support model changeovers.

COMMUNITY INVOLVEMENT

Human Factors and Ergonomics Team Lead

McMaster University Formula SAE Electric Team

September 2025 to Present

- Designed the 2026 vehicle driver interface for a quarter-scale Formula style vehicle in SolidWorks, optimizing ergonomic integration and cross-functionality with the electrical system, while reducing weight by 5%.
- Oversaw multi-component fabrication, including tube notching, and jigging, utilizing 3D-printed fixtures for the head rest and incorporating carbon fiber infusion for seats and dashboard.
- Led a team of 10 engineers, guiding design and manufacturing while managing sponsorship outreach.

Choreographer

McMaster Engineering Musical

October 2025 to Present

- Choreographed dance pieces to support visual storytelling, collaborating closely with the Director and Assistant Director to align movement with overall creative vision.
- Led inclusive rehearsals and weekly reviews, adapting instruction for varied experience levels and fostering a supportive environment that built confidence and technical skill.

Secretary

American Indian Science and Engineering Society (AISES), McMaster University

September 2023 – Present

- Organized chapter meetings, maintained records, and supported event planning and member engagement initiatives.
- Represented the chapter at the 2024 Canadian National Conference (Vancouver), 2025 Canadian National Conference (Toronto), and 2025 American Indian Science and Engineering Society National Conference (Minneapolis).

Internal Liaison

McMaster Indigenous Health Movement

September 2023 to Present

- Consultant for the 2025/26 Executive Teams (Speakers, Community Learning, and Graphics) and responsible for communication between executive and general members,
- Served as Co-Chair for the 2024/25 Executive Team, launching a podcast amplifying Indigenous voices while also managing over 35 executive and general members.
- Led logistical planning and outreach with 5 professional speakers for the 2024 Annual Conference (75+ attendees) that explored identity, advocacy, and interconnectedness.

Welcome Week Representative

Faculty of Engineering, McMaster University

March 2024 to Present

- Supported incoming first-year engineering students through orientation programming, peer mentorship, and inclusive community-building initiatives.
- Co-led an introduction to first-year academics panel, focusing on course expectations and answering frequently asked questions about grading and workload.

Physical Activity Centre of Excellence (PACE) Student Volunteer

McMaster University

January 2024 to January 2025

- Supported clients with physical disabilities during rehabilitation and adaptive activity sessions, assisting with task modification and device use to promote independence.

VP Events

McMaster Society of Mechanical Engineering

September 2023 to Present

- Planned and executed academic, social, and professional events for mechanical engineering students.
- Previously served as VP Internal and coordinated internal operations for 50+ general members and ensure logistical details are thoroughly planned.

McMaster Women in Engineering Society Student Ambassador

September 2023 to April 2025

- Represented Women in Engineering at outreach events and student programming, promoting equity, inclusion, and engagement within the engineering community.
- Conducted bi-annual presentations at St. Patrick High School to talk about post-secondary opportunities in STEM and my experience as an undergraduate engineering student.

McMaster Women in Engineering Society Student Mentor

September 2023 to April 2025

- Mentored junior engineering students through academic transition, community involvement, and professional development.

PROJECT EXPERIENCE

Automated Plant-Watering System

McMaster University

Winter 2025

- Designed an automated fluid-delivery system using an Arduino-controlled peristaltic pump to provide scheduled, species-specific water dosing.
- Developed a Python-based input database and validated flow-rate accuracy by comparing calculated and experimental outputs.
- Recipient of Design Project Distinction Award; project showcased at the Faculty Student Excellence Event.

Assistive Dexterity Device for Earring Backing Application

McMaster University

Winter 2023

- Designed and prototyped a user-centred assistive device for individuals with impaired hand dexterity (MS), focusing on ergonomics, safety, and independence in daily living tasks.
- Applied iterative mechanical design, CAD (Autodesk Inventor), and 3D printing to address human-factors risks and usability constraints; validated through hands-on testing and end-user feedback.
- Coordinated documentation, design reviews, and final demonstration; prototype showcased at the iBioMed Year-End Design Showcase.

Wearable Device for Visual Impaired Swimmers

McMaster University

Winter 2023

- Co-designed a wearable assistive device enabling visually impaired swimmers to navigate pool boundaries independently using distance sensing and non-visual (haptic/audio) feedback.
- Led mechanical design and enclosure development in CAD (Autodesk Inventor), integrating waterproofing, ergonomics, and hydrodynamic considerations, validated functionality through prototype assembly and testing.

SELECTED HONOURS AND AWARDS

2022 Ralph M. Barford Loran Scholar

March 2022

- Valued at \$100,000 and elected based on character, service commitment, and leadership potential; Selected as one of 35 Scholars from 5,174 applicants; Includes tuition waiver, semi-annual stipends, mentorship, and summer internship support.

William Barry Hill Scholarship

September 2025

- \$8,000 award to an Indigenous engineering student with high GPA.

McMaster University Award of Excellence

May 2023

- \$3,000 award to the Top 10% of Engineering Faculty GPAs for iBioMed 1 entries

Dean's Honour List

2022 to Present

- Awarded to students who maintain a sessional average above 85% while maintaining a full-time course load.

Métis Health Policy Forum Youth Representative

February 2025

- Represented Métis youth in Ontario in discussions concerning Métis health inequities, future federal policies related to Métis health, and inter-provincial efforts to strengthen Métis health programming.

Indigenous Healing & Wellness Strategy Youth Panelist

October 2024

- Selected as a youth panelist to share lived experience and perspectives on Indigenous healing, wellness, and community-led health initiatives.